

# SKYLAR JEAN CALLIS

## Curriculum Vitae

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## EDUCATION

University of Delaware

Working Towards MS in Data Science & PhD in Applied Mathematics

Michigan Technological University (MTU)

Summa Cum Laude BS in Civil Engineering | 2017

Summa Cum Laude BS in Applied Mathematics | 2017

## SKILLS:

**Advanced:** Python, Git, Google Suite, Public Speaking, Technical Writing

**Proficient:** Matlab, Wolfram Mathematica, Julia, Linux, LaTeX, Microsoft Office

## ACCOLADES

- First place in the 2020 Airport Cooperative Research Program's University Design Competition in the Runway Safety/Runway Incursions/Runway Excursions Category
- Awarded 2021 Airport Cooperative Research Program's Next Steps Grant
- Awarded 2022 Charles Knobloch Award, 2021 Mathematics Achievement Award, and 2020 Woman of Promise Award by the MTU Mathematical Sciences Department
- Certificates of Merit from the MTU Mathematical Sciences Department in 7 courses:
  - MA3160 (Multivariable Calculus), MA2320 (Intro to Linear Algebra), MA3560 (Differential Equations w/ Modeling), MA3310 (Intro to Abstract Algebra), MA2600 (Scientific Computing), MA4515 (Intro to Partial Differential Equations), & MA4410 (Complex Functions)

## PUBLICATIONS

Hickmann, Kyle, Egozi, Gal, Pandit, Sourabh, Chakrabarti, Sharmistha, Kaiser, Bryan, & Callis, Skylar. Yoke. Computer Software. <https://github.com/lanl/Yoke>. USDOE Office of Science (SC), Advanced Scientific Computing Research (ASCR). 10 Mar. 2025. Web. <https://doi.org/10.11578/dc.20250418.1>.

Callis, S. "Vision Transformers, Explained." *Medium*, Towards Data Science, 27 Feb. 2024, <https://towardsdatascience.com/vision-transformers-explained-a9d07147e4c8>.

Callis, S. "Attention for Vision Transformers, Explained." *Medium*, Towards Data Science, 27 Feb. 2024, <https://towardsdatascience.com/attention-for-vision-transformers-explained-70f83984c673>.

Callis, S. "Position Embeddings for Vision Transformers, Explained." *Medium*, Towards Data Science, 27 Feb. 2024, <https://towardsdatascience.com/position-embeddings-for-vision-transformers-explained-a6f9add341d5>.

Callis, S. "Tokens-to-Token Vision Transformers, Explained." *Medium*, Towards Data Science, 27 Feb. 2024, <https://towardsdatascience.com/tokens-to-token-vision-transformers-explained-2fa4e2002daa>.

Hickmann, K, Callis, S, & Andrews, S. "Training and Interpretability of Deep-Neural Methods for Damage Calibration in Copper." *Proceedings of the ASME 2023 Verification, Validation, and Uncertainty Quantification Symposium*. ASME 2023 Verification, Validation, and Uncertainty Quantification Symposium. Baltimore, Maryland, USA. May 17–19, 2023. V001T04A001. ASME. <https://doi.org/10.1115/VVUQ2023-108759>

Black, J.K., Callis, S.J., Feizy, A., Johnson, C.L., Lieven, N.A.J., & Vega, M.A. (2024). A Simplified Finite Element Joint Model Updated with Experimental Modal Features. In: Allen, M., Blough, J., Mains, M. (eds) Special Topics in Structural Dynamics & Experimental Techniques, Volume 5. SEM 2023. Conference Proceedings of the Society for Experimental Mechanics Series. Springer, Cham. [https://doi.org/10.1007/978-3-031-37007-6\\_12](https://doi.org/10.1007/978-3-031-37007-6_12)

- Anderson, L., **Callis, S.**, Helman, R., & Wehner, K. (2021, December). *The Runway Intersection Marking Project Next Steps* [PDF]. Houghton, MI: Michigan Technological University.
- Anderson, L., **Callis, S.**, Helman, R., & Wehner, K. (2021, December). *An Investigation of Runway Intersection Marking Alternatives: The Runway Intersection Marking Project Next Steps Supplement* [PDF]. Houghton, MI: Michigan Technological University.
- Anderson, L., **Callis, S.**, & Wehner, K. (2020, April). *Runway Intersection Markings: For Increasing Situational Awareness and Decreasing the Frequency of Runway Incursions* [PDF]. Houghton, MI: Michigan Technological University. Retrieved from <https://tinyurl.com/yxzl9aeg>

## CONFERENCES

- “Comparing CNNs & VisTransformers Trained on High Fidelity Simulations” Invited Poster, Salishan Conference on High Speed Computing; Gleneden Beach, OR; April 2024
- “An Overview of Interpretability Methods for Deep Neural Networks as Applied to TePla Damage Calibration in Copper” Presentation, American Society of Mechanical Engineers (ASME) Verification, Validation, and Uncertainty Quantification (VUQ) Symposium; Baltimore, MD; May 2023
- “Interpretability Methods for Deep Neural Networks as Applied to TePla Damage Calibration in Copper” Poster, Conference for Data Analysis (CoDA); Santa Fe, NM; March 2023
- “A Simplified Finite Element Joint Model Updated with Experimental Modal Data” Presentation, Society for Experimental Mechanics (SEM) International Modal Analysis Conference (IMAC); Austin, TX; February 2023
- “Runway Intersection Markings” Invited Presentation, Airport Consultants Council (ACC) Airports Technical Workshop; Online; August 2020

## RESEARCH EXPERIENCE

### GRADUATE RESEARCH ASSISTANT IN

#### APPLIED MATH & PLASMA PHYSICS (June 2025 - Aug. 2025, *seasonal position*)

Los Alamos National Laboratory [LANL] (Los Alamos, NM)

- Integrated into an existing team to develop machine learning model components that classify astronomical transients and complete parameter estimation, with the ultimate goal of creating a foundation model for kilonovae
- Developed machine learning workflows compatible with existing data processing pipelines
- Implemented normalizing flow models using Pyro and Zuko in Python and combined them with existing transformer model components built in PyTorch
- Attended Zwicky Transient Facility (ZTF) Summer School 2025 hosted at the University of Minnesota, where I expanded my astrophysics understanding to improve machine learning model development

#### POST-BACHELORS STUDENT IN COMPUTATIONAL PHYSICS (Oct. 2022 - May 2024)

Los Alamos National Laboratory [LANL] (Los Alamos, NM)

- Implemented neural network sensitivity measures to analyze image-to-scalar regression models, with both python’s Tensorflow/Keras and Pytorch packages
- Developed and published python module to GitHub, including documentation and unit testing
- Implemented and trained Vision Transformer machine learning models for image-to-scalar regression
- Developed Vision Transformer tutorials and published on GitHub and in the *Towards Data Science* blog
- Presented machine learning research in talks and posters at conferences on Data Science, VUQ, and High-Performance Computing

#### COMPUTATIONAL RESEARCH METHODS STUDENT (Feb. - Apr. 2023)

The Computational Research Access NETwork [CRANE] (Virtual Seminar - <https://www.cranephysics.org/>)

- Completed five-week unit on implementing numerical methods in python, including Runge-Kutta, 1D finite difference, and fast fourier transforms
- Completed three-week unit on Monte-Carlo methods, which included implementing MC methods in python to simulate casino games and neutron transport
- Completed four-week unit on signal and image processing, which included writing python code to analyze Langmuir probe and plasma interferometer data

#### LOS ALAMOS DYNAMIC SUMMER SCHOOL [LADSS] (June - Aug. 2022, *seasonal position*)

Los Alamos National Lab (Los Alamos, NM)

- Analyzed reduced-order modeling methodologies for bolted connections in finite element software by comparing to a small-scale physical test
- Implemented a Bayesian Optimization algorithm to update finite element modeling parameters, which integrated a Python algorithm with calls to the finite element software ABAQUS
- Attended weekly lectures with hands-on components on topics such as signal processing, modeling dynamic systems, model validation, and more

#### DIRECTOR'S SUMMER PROGRAM [DSP] (May - Aug. 2021, *seasonal position*)

National Security Agency [NSA] (Fort Meade, MD)

- One of 21 participants chosen from 240 applicants nationwide
- Learned about convolutional neural networks and implemented and trained them for novel classification tasks
- Analyzed data from neural network classifiers to improve their performance
- Read unclassified and classified academic literature in unfamiliar subjects and applied the knowledge to my research project
- Presented research project findings to professional researchers at the Institutes for Defense Analysis (IDA) at Princeton Center for Communications Research (CCR)
- Prepared and delivered briefing to the Director of the National Security Agency on cryptanalytic techniques and results as a result of my research project

#### SECRETARY & FOUNDING MEMBER OF THE BUILT WORLD ENTERPRISE [BWE] (Jan. 2019 - Dec. 2021)

Michigan Technological University [MTU]'s Enterprise Program (Houghton, MI)

- Developed innovative solutions to address existing problems in the field of civil aviation surface safety, and evaluated these solutions with a cross-disciplinary approach
- Integrated research in human factors, saliency, and situational awareness into a civil engineering foundation
- Gathered interdisciplinary feedback from aviation and human factors industry professionals
- Produced technical writing to describe research practices and results, to persuade for the implementation of innovative solutions, and to apply for external funding
- Documented current practices and knowledge of the executive board for future members to reference through the use of meeting minutes and a shared file system
- Lead workshops for undergraduate and graduate students on Reader Expectation Theory as a way to advance their technical writing

## **TEACHING EXPERIENCE**

TEACHING ASSISTANTSHIP (Aug. 2024 - Present)

University of Delaware (Newark, DE)

- MATH241 (Analytic Geometry and Calculus A), Fall 2024
- MATH241 (Analytic Geometry and Calculus A), Spring 2025
- MATH010 (Intermediate Algebra), Fall 2025
- MATH231 (Integrated Calculus IA), Fall 2025